Name: Aadya Arvind

SRN: PES2UG24CS008

Section: 5A

LAB 7: FUZZ TESTING FOR INPUT VALIDATION BUGS

Hypothesis-based testing and input validation

Evidence screenshots

This document contains the execution evidence for the original application, the failing Hypothesis tests, the corrected application, and the final passing tests.

Screenshot 1 – Original application execution

The original processor.py was executed with valid sample inputs for a promo voucher, seat tier, and attendee badge.

```
C:\Users\Aadya\L6_L7_06_event_ticket_tier\Lab 7>py processor.py
==== Running decode_promo_voucher ====
Enter voucher (e.g. EARLY-20): EARLY-20
Voucher: Valid EARLY: 20% OFF

==== Running calculate_seat_tier_price ====
Enter tier spec (e.g. VIP@4): VIP@4
Seat Price: $ 158.0

==== Running parse_attendee_badge ====
Enter badge CSV (Name,Age,Seat): Aadya,20,A12
Badge: Aadya [Adult] - Seat A12
```

Screenshot 2 – Hypothesis tests before fixing

All three property-based tests failed against the original implementation, demonstrating that unexpected inputs can cause failures.

```
C:\Users\Aadya\L6_L7_06_event_ticket_tier\Lab 7>py -m pytest test_processor.py -v
============================================================ test session starts ============================================================
platform win32 -- Python 3.12.0, pytest-8.3.4, pluggy-1.5.0 -- C:\Program Files\Python312\python.exe
cachedir: .pytest_cache
hypothesis profile 'default'
rootdir: C:\Users\Aadya\L6_L7_06_event_ticket_tier\Lab 7
plugins: anyio-4.6.0, hypothesis-6.170.15
collected 3 items

test_processor.py::test_decode_promo_voucher_no_crash FAILED                                                                        [ 33%]
test_processor.py::test_calculate_seat_tier_price_safe FAILED                                                                       [ 66%]
test_processor.py::test_parse_attendee_badge_safe FAILED                                                                            [100%]
```

Screenshot 3 – Fixed application execution

The corrected processor.py was executed with the same valid sample inputs to confirm that expected valid-input behaviour is preserved.

```
C:\Users\Aadya\L6_L7_06_event_ticket_tier\Lab 7>py processor.py
==== Running decode_promo_voucher ====
Enter voucher (e.g. EARLY-20): EARLY-20
Voucher: Valid EARLY: 20% OFF

==== Running calculate_seat_tier_price ====
Enter tier spec (e.g. VIP@4): VIP@4
Seat Price: $ 158.0

==== Running parse_attendee_badge ====
Enter badge CSV (Name,Age,Seat): Aadya,20,A12
Badge: Aadya [Adult] - Seat A12
```

Screenshot 4 – Hypothesis tests after fixing

All three Hypothesis tests passed after defensive input validation was added to the three functions.

```
C:\Users\Aadya\L6_L7_06_event_ticket_tier\Lab 7>py -m pytest test_processor.py -v
============================= test session starts =============================
platform win32 -- Python 3.12.0, pytest-8.3.4, pluggy-1.5.0 -- C:\Program Files\Python312\python.exe
cachedir: .pytest_cache
hypothesis profile 'default'
rootdir: C:\Users\Aadya\L6_L7_06_event_ticket_tier\Lab 7
plugins: anyio-4.6.0, hypothesis-6.170.15
collected 3 items

test_processor.py::test_decode_promo_voucher_no_crash PASSED             [ 33%]
test_processor.py::test_calculate_seat_tier_price_safe PASSED            [ 66%]
test_processor.py::test_parse_attendee_badge_safe PASSED                 [100%]

============================== 3 passed in 1.93s ==============================
```